



Control of stress-induced depressive disorders by So-ochim-tang-gamibang, a Korean herbal medicine



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Liquiritin (PubChem CID: 503737)

Naringin (PubChem CID: 442428)

Platycodin D (PubChem CID: 162859)

Linderane (PubChem CID: 5386162)

Costunolide (PubChem CID: 5281437)

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ABSTRACT

Ethnopharmacological relevance: So-ochim-tang-gamibang (SOCG) is a Korean herbal medicine formula that has been applied to treat depressive moods and depression associated somatoform pain. This decoction consists of *Cyperus rotundus* L. (Cyperus Rhizoma), *Lindera aggregata* (Sims) Kosterm. (Linderae Radix), *Aquilaria agallochum* (Lour.) Roxb. ex Finl. (Aquilariae Resinatum Lignum), *Glycyrrhiza uralensis* Fisch. (Glycyrrhizae Radix) *Platycodon grandiflorum* (Jacq.) A. DC. (Platycodi Radix), and *Citrus aurantium* L. (Aurantii Fructus). The aim of this study is to assess antidepressant-like effects of SOCG and to investigate its possible cellular and molecular mechanisms.

Material and methods: Using chronic restraint stress animal model, effects of SOCG on depressive-like behaviors, corticosterone, and hippocampal expressions of a neurotrophic factor and an apoptotic marker, were investigated. Mice were exposed to restraint stress 6 h per day over a period of two weeks, and orally administrated either SOCG (30, 100, or 300 mg/kg/day). The depressive-like behaviors were analyzed by forced swimming test and open field test. The serum levels of corticosterone were measured by enzyme-linked immunosorbent assay. Expressions of caspase-3 and BDNF in the hippocampus were analyzed by immunofluorescence. Further, effects of SOCG were examined in corticosterone-treated PC12 cells. Cellular toxicity was determined by 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide and lactate dehydrogenase assays. Real-time PCR was applied to investigate the cellular expression levels of Bax, Bcl-2, and BDNF. The levels of caspase-3 and BDNF were examined by Western blotting.

Results: Administration of SOCG not only reduced immobility time of restraint-stressed mice in a dose-dependent manner, but also significantly increased the distance mice moved and the number of crossings in the open field test. Further, SOCG significantly reduced the serum level of corticosterone and expression of caspase-3, while increased expression of BDNF *in vivo*. SOCG increased cell viability in corticosterone treated PC12 cells, which was accompanied by decreased caspase-3 expression and the ratio of Bax/Bcl-2 mRNA expression as well as increased BDNF expression *in vitro*.

Conclusions: Taken together, our data suggested that SOCG may have potential as an antidepressant agent controlling depressive behaviors and corticosterone-induced neuronal damage caused by chronic stress.

Abbreviations: SOCG, So-ochim-tang-gamibang; BDNF, brain-derived nerve growth factor; HPA, hypothalamic-pituitary-adrenal; ACTH, adrenal corticotrophin hormone; FST, forced swimming test; OFT, open field test; LDH, Lactate dehydrogenase; GAPDH, glyceraldehyde-3 phosphate dehydrogenase

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1. Introduction

Major depression is a common psychiatric disorder with an increasing prevalence worldwide (Lepine and Briley, 2011). Clinically, major depression is characterized by a depressed mood in combination with additional symptoms such as loss of appetite and body weight, sleep disturbances, psychomotor agitation and/or retardation, fatigue, feelings of worthlessness or guilt, diminished cognitive functioning, and/or recurrent thoughts of death (Maurer, 2012). The connection between stressful experiences and the onset of a major depressive episode and anxiety has been reported in many previous studies (Finlay-Jones and Brown, 1981; Hettema et al., 2006; Kendler et al., 1998; Siegrist, 2008), and chronic anxiety caused by repetitive stresses has been thought to contribute to the development of depression (Hirschfeld, 2001). Although the pathophysiology of these stress-related disorders is not been completely understood, stress-induced dysfunction of the hypothalamic-pituitary-adrenal (HPA) axis has been known to be associated with the development of depressive symptoms (Pariante and Lightman, 2008) as well as anxiety (Faravelli et al., 2012). The HPA axis is a neuroendocrine system involved in the production of the stress hormone cortisol. Specifically, via the HPA axis, corticotrophin-releasing hormone is secreted from the hypothalamus in response to perceived psychological stress, which induces production and peripheral circulation of adrenal corticotrophin hormone (ACTH) from the pituitary gland, and ACTH promotes secretion of cortisol from the adrenal gland into the blood. Homeostasis of the HPA axis is maintained by feedback inhibition of cortisol. However, progressive HPA axis abnormalities caused by stress can lead to high circulating levels of cortisol, which not only triggers depression- and anxiety-like changes in behavior (Gregus et al., 2005; Nemeroff and Vale, 2005; Pariante and Lightman, 2008) but also induces changes in brain structure due to decreased cell proliferation and neuronal cell loss. Cortisol shows direct toxic effects on the brain (Herbert et al., 2006) and attenuates or reduces neurotrophic factors including brain-derived nerve growth factor (BDNF). Indeed, the average concentration of cortisol is reportedly higher in depressed patients than in healthy control subjects (Gorwood et al., 2008). Additionally, the hippocampal shrinkage commonly observed in depressed patients has been linked to suppressed neurogenesis and neuronal toxic damage in association with elevated levels of cortisol (Hoschl and Hajek, 2001) and reduced expression of BDNF (Murakami et al., 2005).

So-ochim-tang is a prescription that is described in the qi chapter of Donguibogam, a well-documented traditional textbook of Korean medicine (UNESCO, 2009). Qi stagnation was suggested to be the cause of depression in traditional Korean and Chinese medicine (Yeung et al., 2014). So-ochim-tang has been used to treat depressive moods and somatoform pain by relieving qi stagnation. It consists of *Cyperus rotundus* L. (Cyperus Rhizoma), *Lindera aggregata* (Sims) Kosterm. (Linderae Radix), *Aquilaria agallocha* (Lour.) Roxb. ex Finl. (Aquilariae Resinatum Lignum), and *Glycyrrhiza uralensis* Fisch. (Glycyrrhizae Radix). Based on Korean medical theory, So-ochim-tang-gamibang (SOCG), which has been used in clinical practice for the last two decades, is formulated from So-ochim-tang by adding *Platycodon grandiflorum* (Jacq.) A. DC. (Platycodi Radix) and *Citrus aurantium* L. (Aurantii Fructus). These additives are suggested to enhance the efficacy by increasing qi circulation (Lee et al., 2010; Ryu et al., 2008). *Aquilariae Resinatum Lignum* in the original formula is substituted for *Aucklandia lappa* DC. (*Aucklandia Radix*) in the preparation of SOCG. These plants are regarded as complementary or as substances that can be used interchangeably as substitutes for each other, with a similar efficacy in Korean medicine (Hwang, 2007). We previously reported that SOCG suppressed the activities of serotonin and protected cells from oxidant stress *in vitro*, suggesting the antidepressant effect of the decoction (Hwang et al., 2013). Iridoid glycosides and cycloartane from *C. rotundus* were recently suggested to have anti-depressant activity (Zhou et al., 2016a, 2016b). Liquiritin

and isoliquiritin from *G. uralensis* were reported to suppress depression-like behavior in mice by increasing 5-HT and norepinephrine (Wang et al., 2008). Anxiolytic activity was also observed in *C. aurantium*-treated mice (Costa et al., 2013). In this study, we assessed the effects of SOCG and its underlying mechanisms on restraint stress-induced depression- and anxiety-like behaviors and corticosterone-induced neuronal damage in mice.

2. Material and methods

2.1. Animals

Male C57/BL6 mice (7 weeks old, 18–22 g) were purchased from DBL (Daehan Bio Link, Chung-buk, Korea). Mice were housed in pairs and maintained using standard conditions (12:12 h light-dark cycle, 23 ± 1°C) with free access to water and food. The University Animal Care and Use Committee at Daejeon University approved all of the animal experimental protocols (DJUAR2014-030).

2.2. Preparation of SOCG water extracts and drug

The six crude herbs, *Cyperus Rhizoma* (*Cyperus rotundus* L.), *Lindera Radix* (*Lindera strychnifolia* Fern.-Vill.), *Aucklandia Radix* (*Aucklandia lappa* Decne.), *Glycyrrhizae Radix* (*Glycyrrhiza uralensis* Fisch.), *Aurantii Fructus* (*Citrus aurantium* L.), and *Platycodi Radix* (*Platycodon grandiflorus* Jacq. A.DC) were purchased from Dong Kyung Pharmaceutical Co., Ltd. (Seoul, Korea). SOCG extract was obtained by boiling the six herbs (ratio 8: 4: 1: 1: 4: 4, in the order given above) in distilled water at 100°C for 2 h. The volume of distilled water was 10 times the total herb weight (110 kg/1,100 L, w/v). This decoction was evaporated under reduced pressure at 60 °C for 5 h and freeze-dried. The drug:extract ratio (DER) was 6.8 (yield ratio, 14.6%). The extracted SOCG powder was stored at –20°C. Voucher specimens (No. 194A079-85) of collected herb samples were deposited in the herbarium of Han Kook Shin Yak Co., Ltd. (Nonsan, Korea). Fluoxetine was purchased from Sigma-Aldrich Co. (St Louis, MO, USA).

2.3. High-performance liquid chromatography (HPLC) analysis of SOCG

To determine the chemical profile of SOCG, 20 mg of SOCG was dissolved in 1 mL of methanol followed by membrane filtration (0.45 µm). Three-dimensional HPLC analysis was performed using a Waters system (Waters Corp., Milford, MA, USA) consisting of a separation module (e2695) with a photodiode array detector (2998). UV absorbance was monitored from 190 to 500 nm. Qualitative analysis was conducted at 210 nm and column temperature was maintained at 30°C. The separation was performed using a YMC-Triart C18 column (250×4.6 mm; particle size, 5 µm; YMC Co. Ltd., Kyoto, Japan). The mobile phase comprised 0.1% phosphoric acid-water (v/v, solvent A) and acetonitrile (solvent B). The flow rate was 1 mL/min, and the gradient was as follows: 0 min, 85% A; 20 min, 60% A; 25 min, 35% A; 40 min, 20% A. The re-equilibration time was set at 20 min.

2.4. Restraint stress and administration of SOCG

All mice were randomly divided into six groups (n=6 or 8/group) as follows: normal group, control group, SOCG groups (30, 100, or 300 mg/kg), and fluoxetine (a well-known antidepressant) group (20 mg/kg). To produce restraint stress, with the exception of those in the normal group, mice were individually placed in a perforated Falcon tube for 6 h per day (from 11:00 to 17:00 h) every day for 2 weeks. For exposure to the tested substances, 2 h before restraint, SOCG (30, 100, or 300 mg/kg) and fluoxetine (20 mg/kg) were orally administered with distilled water as the vehicle using a 22-gauge feeding needle (0.1 mL/10 g body weight).

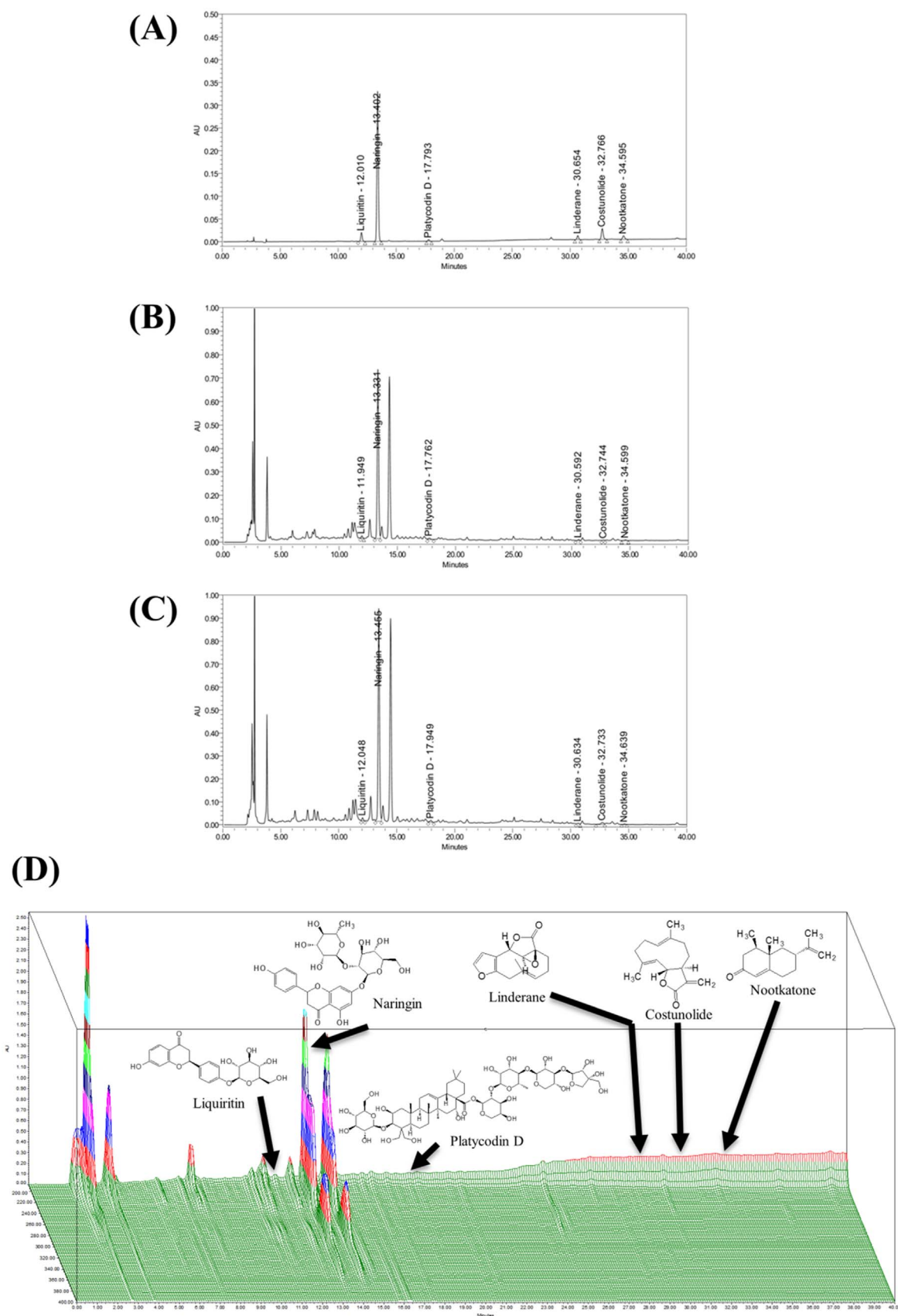


Fig. 1. HPLC chromatogram for the determination of liquiritin, naringin, platycodin D, linderane, costunolide, and nootkatone in SOCG. Chromatogram of (A) six standards, (B) SOCG batch I, and (C) SOCG batch II. (D) 3D-HPLC profile of SOCG. Chemical structure of each standard compound is presented.

Table 1

Contents (μg/mg) of six standards in SOCG.

Herbal name	Scientific name	Local name	Reference compound	r ²	Content
Glycyrrhizae Radix	<i>Glycyrrhiza uralensis</i> Fisch.	Gamcho (甘草)	Liquiritin	0.9999	0.442 ± 0.008
Aurantii Fructus	<i>Citrus aurantium</i> L.	Zigak (枳殼)	Naringin	0.9964	14.584 ± 0.083
Platycodi Radix	<i>Platycodon grandiflorum</i> (Jacq.) A. DC.	Gilgyeong (桔梗)	Platycodin D	0.9999	2.913 ± 0.013
Linderae Radix	<i>Lindera aggregata</i> (Sims) Kosterm.	Oyak (烏藥)	Linderane	0.9963	0.075 ± 0.002
Aucklandiae Radix	<i>Aucklandia lappa</i> DC.	Mokhyang (木香)	Costunolide	0.9999	0.004 ± 0.001
Cyperi Rhizoma	<i>Cyperus rotundus</i> L.	Hyangbuja (香附子)	Nootkatone	0.9993	0.035 ± 0.001

2.5. Behavioral tests

After the repeated restraint stress procedure, the mice were challenged in the forced swimming test (FST) and open field test (OFT). All tests were performed between 09:00 and 14:00 h in a quiet room. For the FST, mice were individually forced to swim in a cylinder (46 cm height, 20 cm diameter) containing tap water (25 ± 2°C, 20 cm depth). Total immobility time was measured during the last 4 min of a 6 min swim using video tracking software (SMART 3.0; Panlab S.I., Barcelona, Spain). The ambulatory behavior of mice was assessed in the OFT. The open field arena (30 cm×30 cm) was constructed from acrylic sheets, and each arena was separated into four zones. Their behavior was observed and videotaped for 10 min using the same software as in the FST. Travel distance and the number of crossings were used as the parameters.

2.6. Corticosterone assay

One day after the behavioral tests finished, the blood was collected from the hearts of sacrificed mice under ketamine anesthesia. Corticosterone levels in serum were measured using a competitive enzyme-linked immunosorbent assay (ELISA) kit (Enzo life sciences, NY, USA) according to the manufacturer's protocols.

2.7. Brain tissue extraction and immunohistochemistry assay for BDNF and caspase-3 expression

On day 15, brain tissues were removed from sacrificed mice and immediately frozen at −75°C. The frozen tissue samples were cut into 20 μm slices using a cryostat and mounted on positively charged slides (Fisher Scientific, Pittsburgh, PA, USA). Immunohistochemistry was conducted using previously described methods (Han et al., 2007). The primary antibodies used were anti-BDNF (rabbit polyclonal; Santa Cruz Biotechnology Inc., Dallas, TX, USA), anti-caspase-3 (rabbit polyclonal; Cell Signaling Technology Inc., Danvers, MA, USA) antibodies, and the secondary antibody was rhodamine-goat anti-rabbit antibody (Molecular Probes, Rockford, IL, USA). Then, the nuclei were stained with Hoechst 33258 (Sigma-Aldrich, St. Louis, MO, USA) for 10 min, and the fluorescence intensity was analyzed using the ImageJ software (National Institutes of Health, NIH, <http://imagej.nih.gov/ij/>, Version 1.49).

2.8. Cell culture

PC12 cells were purchased from Korean Collection for Type Cultures (Daejeon, Korea). They were cultured in RPMI medium supplemented with penicillin (100 unit/mL), streptomycin (100 μg/mL), 5% fetal bovine serum, and 10% horse serum at 37°C in a humidified atmosphere containing 5% CO₂.

2.9. 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) assay

Cells were seeded into 96-well plates at a density of 10⁵ cells/mL and left for 24 h prior to being cultured in a serum-free medium for

4 h. Subsequently, the cells were pre-incubated with or without SOCG for 1 h followed by incubation with 250 μM corticosterone for 24 h. Cells were treated with the MTT solution for 3 h. The formazan crystals were solubilized with DMSO. After 30 min incubation, absorbance at 570 nm was measured with a microplate reader (Molecular Device, Sunnyvale, CA, USA).

2.10. Lactate dehydrogenase (LDH) assay

LDH assays were performed using the supernatants and an LDH assay kit (Takara Bio, Shiga, Japan) according to the manufacturer's protocol. Briefly, after treatment PC12 cells were incubated for 6 h, centrifuged for 10 min, and then the supernatants were collected (100 μL/well). In the LDH assay, the supernatants were mixed with a prepared 100 μL reaction mixture for 30 min and absorbance at 490 nm was measured with a microplate reader (Molecular Devices).

2.11. Western blot analysis

PC12 cells were plated at a density of 10⁶ cells/well in a 6-well plate. SOCG was pretreated for 1 h, and then the cells were incubated with corticosterone for 24 h. The cells were collected and lysed and then boiled for 5 min. To separate proteins, electrophoresis was conducted, and they were transferred to PVDF membranes. The anti-caspase-3 and BDNF antibodies (1:1000; Santa Cruz Biotechnology Inc.) were applied during an overnight incubation at 4°C. Finally, membranes were incubated with horseradish peroxidase-labeled anti-rabbit or anti-mouse IgG (1:2000; Santa Cruz Biotechnology Inc.) for 1 h at room temperature. At the level of detection, the analysis was performed using the ECL Western Blotting Detection System (Amersham Biosciences, Pittsburgh, PA, USA). The band intensities were measured using the ImageJ software (Version 1.49).

2.12. Real-time PCR

PC12 cells were seeded at a density of 10⁶ cells/well in a 6-well plate. The cells were pretreated with SOCG for 1 h followed by incubation with corticosterone for 24 h or 48 h for measuring of Bax and Bcl-2 mRNA expression or BDNF mRNA expression, respectively. Total RNA was isolated by using Trizol reagent (Invitrogen, Carlsbad, CA, USA) and then used for cDNA synthesis, which was conducted with the PrimeScript™RT reagent kit (TaKaRa). The mRNA of Bax, Bcl-2, BDNF, and glyceraldehyde-3 phosphate dehydrogenase (GAPDH) was quantified using a 7500 real-time PCR system (Applied Biosystems, CA, U.S.A.) with Power SYBR®Green PCR Master Mix and TaqMan®Gene Expression Master Mix (Applied Biosystems, Foster City, CA, USA). The sequences of the real-time PCR primers were as follows: rat Bax forward: 5'- ACACCTGAGCTGACCTTG -3', reverse: 5'- AGCCCATGATGGTTCTGATC -3'; rat Bcl-2 forward: 5'- CATGC GACCTCTGTTTGA -3', reverse: 5'- GTTTCATGGTCCATCCTTG -3'; rat BDNF forward: 5'- CAGCTGGGTAGGCCAAGTTG -3', reverse: 5'- CACAATGTTCACCAGGTGAGA -3'; and rat GAPDH primer set: Endogenous Control (VIC®/MGB Probe, Primer Limited) from Applied Biosystems (4352338E). The PCR reaction was run for 40 cycles at 95°C (15 s) and 60°C (1 min).

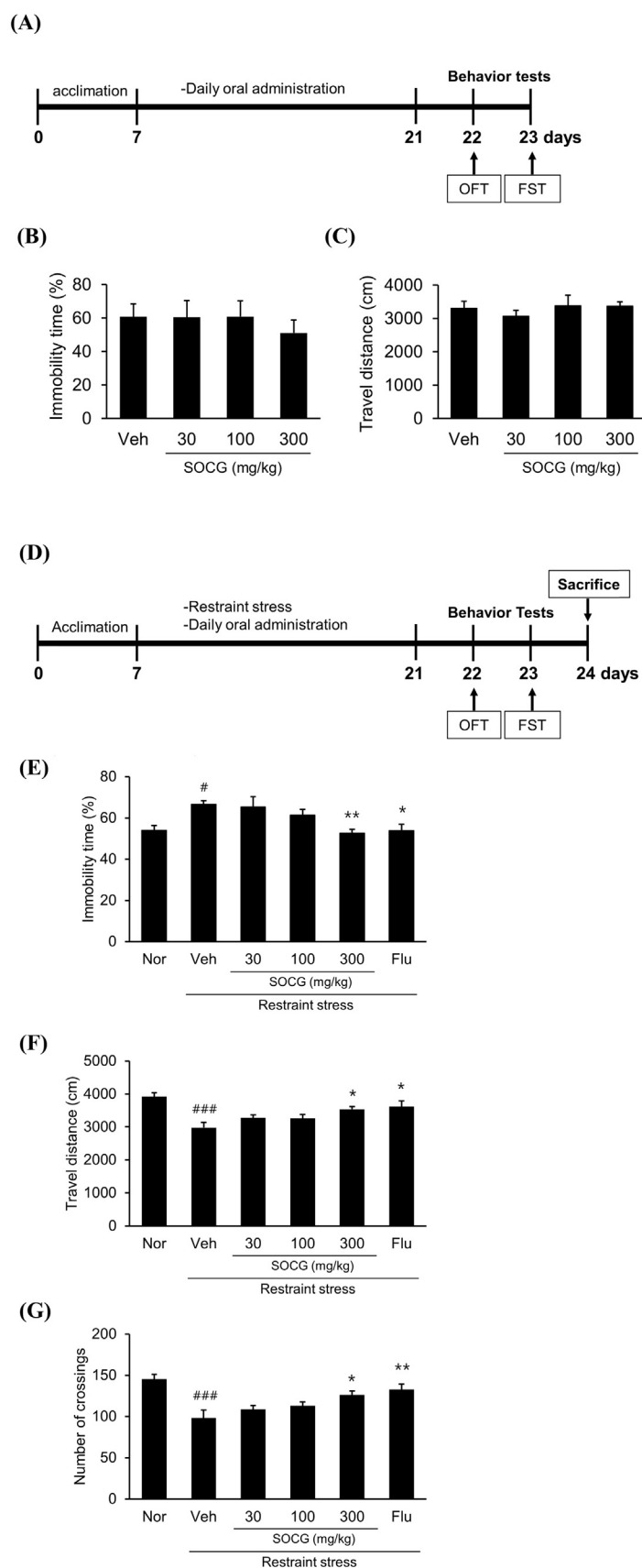


Fig. 2. Effects of SOCG on depression- and anxiety-like behaviors in restraint-stressed mice. (A) Schematic diagram of experimental procedure for SOCG treatment (n=4/group) in normal condition. Mice were administered vehicle (distilled water) or SOCG (30, 100, and 300 mg/kg) for 14 days followed by performance of (B) FST and (C) OFT. (D) Schematic diagram of experimental procedure for SOCG treatment in restraint-stressed mice (n=6 or 8/group). Mice were administered vehicle (distilled water), SOCG (30, 100, and 300 mg/kg), or fluoxetine (20 mg/kg). (E) Immobility time in FST. (F) Travel distance and (G) number of crossings in OFT. Data are mean $\pm$ SEM. * $p < 0.05$, *** $p < 0.001$ significantly different from normal group. * $p < 0.05$, ** $p < 0.01$ significantly different from vehicle group. Nor, normal; Veh, vehicle (distilled water); Flu, fluoxetine; FST, forced swimming test; OFT, open field test.

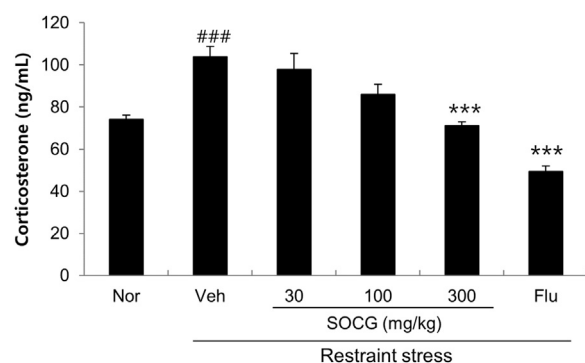


Fig. 3. Effect of SOCG on the levels of corticosterone in serum ($n=6/\text{group}$). Mice were administered vehicle (distilled water), SOCG (30, 100, and 300 mg/kg), or fluoxetine (20 mg/kg). Data are mean $\pm$ SEM. *** $p < 0.001$ significantly different from normal group. *** $p < 0.001$ significantly different from vehicle group. Nor, normal; Veh, vehicle (distilled water); Flu, fluoxetine.

2.13. Statistical analysis

Data were given as mean $\pm$ SEM in the *in vivo* test and mean $\pm$ SD in the *in vitro* test. Multiple group comparisons were performed using one-way ANOVA, followed by post hoc Tukey tests. P values < 0.05 were considered statistically significant.

3. Results

3.1. Standardization of SOCG

For quality control of SOCG, we confirmed the contents of six compounds as surrogate markers using the HPLC. Fig. 1A shows a chromatogram obtained from a standard mixture. The content of each compound in SOCG is shown in Table 1. SOCG is a polyherbal formulation, and the analysis of its composition posed a challenge. Therefore, we subjected two different batches of SOCG extracts (I and II) to HPLC analysis under the same conditions. The linearity of each compound was calculated using three concentrations. We detected identical chromatographic peaks in both batches I and II, indicating an excellent reproducibility of SOCG chemical composition (Fig. 1B and C). The chemical profiles obtained from the standard mixture and SOCG using 3D-HPLC analysis are shown in Fig. 1D. Each SOCG peak was identified by comparing it with the retention times and UV spectra of the standards.

3.2. Effects of SOCG on depressive disorders in restraint-stressed mice

We first tested the effects of SOCG on animal locomotion to exclude the possibility that the medicine induces spontaneous motor activity (Fig. 2A). As shown Fig. 2B and C, the results demonstrated that there was no difference in immobility duration in FST [$F(3, 12)=0.242$, $p=0.865$] (Fig. 2B), and the distance traveled in OFT [$F(3, 12)=0.395$, $p=0.759$] (Fig. 2C) between control and all the SOCG-treated groups. There were no significant differences between all groups following the ANOVA analysis. The results suggested that SOCG did not increase the spontaneous locomotion of the mice.

Next, we assessed the antidepressant effects of SOCG on restraint-stressed mice (Fig. 2D). As shown in Fig. 2E, compared with the normal (unstressed) group, chronic restraint stress increased the immobility of mice when they performed the FST. However, in mice

treated with 300 mg/kg SOCG, immobility time significantly decreased [$F(5, 36)=5.418$, $p < 0.01$]. Furthermore, we investigated the effect of SOCG on depression- as well as anxiety-like behaviors using OFT. Repetitive restraint stress also induced significantly decreased travel distances of mice in the field (Fig. 2F); however, SOCG treatment significantly increased the travel distances of stressed mice, and the difference was significant with a dose of 300 mg/kg [$F(5, 36)=6.392$, $p < 0.05$]. As shown in Fig. 2G, the SOCG treatment groups exhibited an increased numbers of crossings compared to the stressed group, and this increase was significant at 300 mg/kg [$F(5, 36)=7.389$, $p < 0.05$]. Additionally, in mice treated with fluoxetine (20 mg/kg), immobility time in the FST decreased [$F(5, 36)=5.418$, $p < 0.05$], whereas distance traveled decreased [$F(5, 36)=6.392$, $p < 0.05$] and the number of crossings in the OFT increased [$F(5, 36)=7.389$, $p < 0.01$], which are comparable results to those obtained using 300 mg/kg SOCG.

3.3. Effects of SOCG on the levels of corticosterone in mouse serum

As expected, serum corticosterone levels were significantly higher in restraint-stressed mice (103.81 ± 5.06) than in normal unstressed mice (74.13 ± 1.95) [$F(5, 30)=19.806$, $p < 0.001$]. SOCG at 100 (86.01 ± 4.82) or 300 mg/kg (71.23 ± 1.71) reduced corticosterone levels, but the difference was only significant at 300 mg/kg [$F(5, 30)=19.806$, $p < 0.001$]. The mice treated with fluoxetine (49.48 ± 2.48) also showed low levels of corticosterone [$F(5, 30)=19.806$, $p < 0.001$], which were comparable to those of SOCG-treated mice (300 mg/kg) (Fig. 3).

3.4. Effects of SOCG on caspase-3 and BDNF expression in the hippocampus

As shown in Fig. 4, the expression of caspase-3 in the CA3 region was low in the normal group, but in the hippocampal cells of restraint-stressed mice expression of caspase-3 increased. However, in mice treated with SOCG at 300 mg/kg, the expression of caspase-3 was significantly diminished. Expression of BDNF in the normal and stressed control groups was comparable in the CA1 and CA3 region of the hippocampus in stressed mice. When SOCG was administered at 300 mg/kg, BDNF expression increased in the CA1 and CA3 regions.

3.5. Effects of SOCG on corticosterone-induced cell damage

To understand the effects of SOCG on depression at the cellular and molecular levels, we examined the impact of SOCG on corticosterone-induced neuronal cell damage. First, we tested whether SOCG treatment affected cell viability. As shown in Table 2, SOCG did not produce toxic effects in PC12 cells up to 100 $\mu\text{g}/\text{mL}$. In contrast, corticosterone at 250 μM reduced cell viability by 71.9% compared to the viability of normal cells (Table 3). However, pretreatment with SOCG significantly increased cell viability, and the maximum effect was observed at 10 $\mu\text{g}/\text{mL}$ (Table 3). The data from LDH assay indicated that SOCG treatment suppressed the release of LDH in a dose-dependent manner, indicating that SOCG may control cell injury (Table 4). Therefore, we further examined the expression of the cell death-related markers caspase-3, Bax, and Bcl-2. Both expression of caspase-3 and the ratio of Bax to Bcl-2 mRNA expression significantly increased in corticosterone-treated PC12 cells compared with normal cells (Fig. 5A and B). However, SOCG significantly reduced caspase-3 expression and the Bax/Bcl-2 ratio. Corticosterone treatment suppressed the mRNA expression of BDNF, but pretreatment with SOCG significantly increased BDNF expression levels. The peak expression of BDNF was achieved more than 2-fold higher than that of the control (Fig. 5C). The

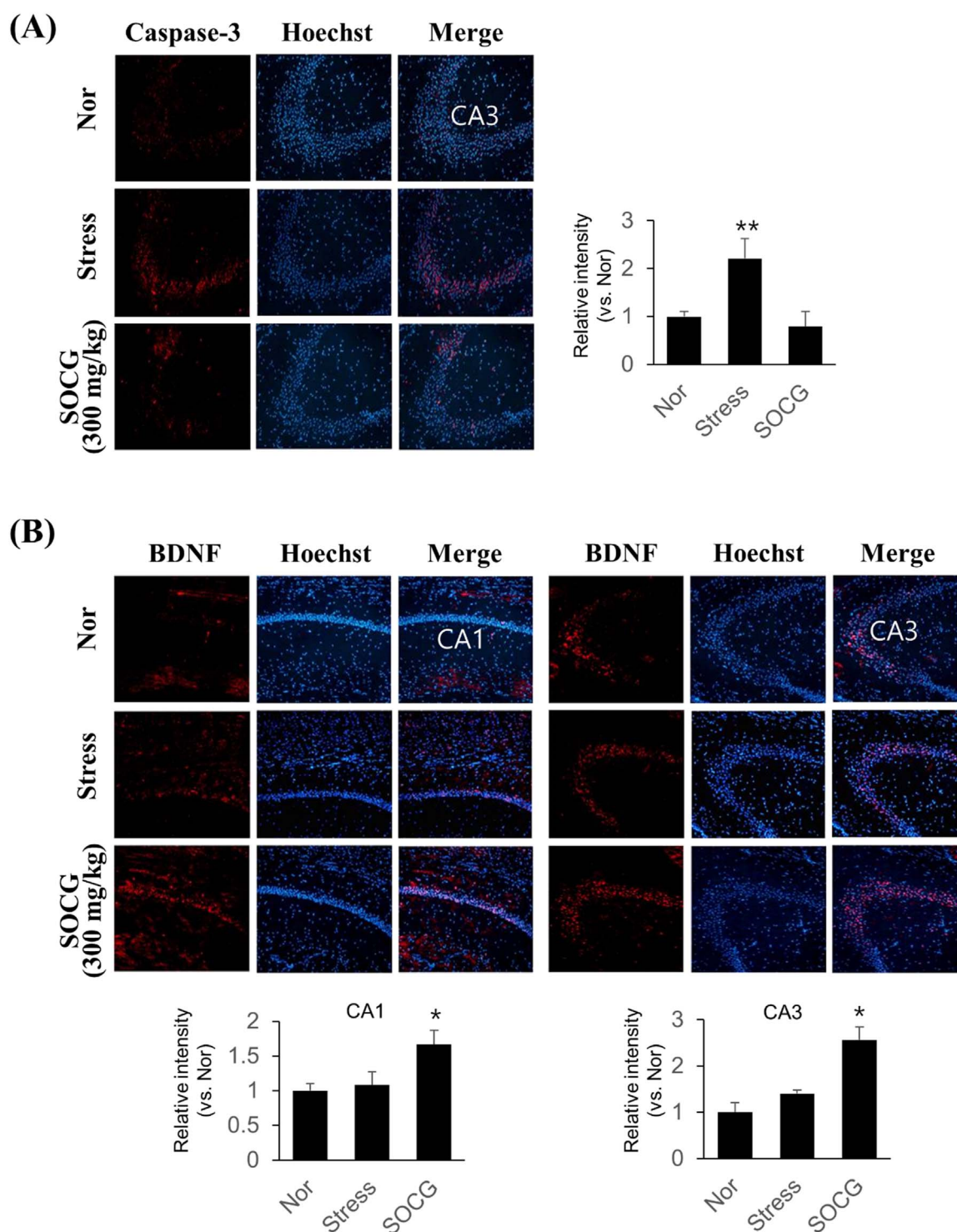


Fig. 4. Effects of SOCG on caspase-3 and BDNF expression in the hippocampus. (A) caspase-3 expression in CA3 region. (B) BDNF expression in CA1 and CA3 region. Data are mean $\pm$ SEM. * $p < 0.05$, ** $p < 0.01$ compared to normal control group. Nor, normal group; stress, restraint stress group.

protein levels of BDNF was also increased by SOCG in corticosterone-treated PC12 cells (Fig. 5D). These results indicate that SOCG might alleviate corticosterone-induced cell damages.

4. Discussion

In this study, we demonstrated that SOCG administration alleviates depression- or anxiety-related behaviors in an animal model. Chronic restraint stress resulted in significantly lowered body weight (data not shown) and increased serum levels of corticosterone in mice, indicating

that the subjects were stressed (Marin et al., 2007; McGill et al., 2006). Productions of caspase-3 and BDNF were also altered in the hippocampus of stressed vs. unstressed mice. Immobilization and reduced activity in FST and OFT are suggested as assessment indicators of depression- and anxiety-like behaviors in stressed mice.

Prolonged exposure to corticosterone has previously been shown to result in brain damage, including attenuation of neurogenesis and promotion of apoptosis (Li et al., 2013; Xu et al., 2004), which contributes to the development of depression. The hippocampus, one of the several regions where neurogenesis occurs, is especially affected.

Table 2

Effect of SOCG on the cell viability in PC12 cells.

Group	Concentration (μg/mL)	Cell viability (%)
Normal	–	100.0 ± 4.89
SOCG	0.1	91.9 ± 2.39
	1	93.8 ± 10.5
	10	97.5 ± 7.84
	100	90.7 ± 2.36
	1000	38.7 ± 0.56 ^{***}

Values are expressed as means ± SD (n=3).

^{***} p < 0.001 significantly different from normal group.**Table 3**

Effect of SOCG on the cell viability in corticosterone-treated PC12 cells.

Group	SOCG (μg/mL)	Corticosterone (250 μM)	Cell viability (%)
Normal	–	–	100.0 ± 19.9
Control	–	+	71.9 ± 11.3 [#]
SOCG	0.1	+	51.9 ± 4.02 [*]
	1	+	78.0 ± 5.58
	10	+	117 ± 24.7 ^{**}
	100	+	99.5 ± 10.3 ^{**}
	1000	+	39.1 ± 1.38 [*]

Values are expressed as means ± SD (n=3).

^{*} p < 0.05 significantly different from corticosterone-treated control group.^{**} p < 0.01 significantly different from corticosterone-treated control group.[#] p < 0.01 significantly different from normal group.

In the hippocampus of patients with major depression, functional defects with altered morphology, atrophy, and/or neuronal cell loss have been reported, as well as a reduction in hippocampal volume (Campbell et al., 2004). Aberrant expression of caspase-3, a proapoptotic protein has been reported in a range of stress-induced depression models (Bachis et al., 2008; Kubera et al., 2011). Expression of BDNF, one of the key molecules for modulating brain plasticity (Chen et al., 2005; Cotman and Berchtold, 2002), is also reportedly diminished or attenuated in corticosterone-injected mice (Gourley et al., 2009; Huang et al., 2011; Jacobsen and Mork, 2006) and depressed animals (Liu and Zhou, 2012; Zheng et al., 2006). Based on these previous studies, we assume that the suppressive effects of SOCG on depression- and anxiety-related behaviors might be primarily mediated by inhibition of corticosterone production, although we do not exclude other possible mechanisms.

Using corticosterone-treated PC12 cells, a pheochromocytoma cell line with high levels of glucocorticoid receptor expression, we showed that SOCG may have the potential to prevent the stress-induced neural cell damage *in vitro*. Induced by corticosterone, the caspases, a family of cysteine proteases, play an essential role in apoptosis. Activation of caspase-3, a final target in the signaling pathway leads to apoptosis by promoting nuclear breakdown and cellular degradation (Budihardjo et al., 1999). Members of the Bcl-2 family regulate activation of caspases by balancing the expression of anti- to pro-apoptotic proteins (Laane et al., 2007). Bax, a member of the Bcl-2 gene family, promotes apoptosis by binding to and antagonizing an anti-apoptotic Bcl-2 protein in corticosterone-treated cells (Renault et al., 2013; Schlossmacher et al., 2011). In our study, SOCG was not toxic to PC12 cells even at high concentration (100 μg/mL) (Table 2). In contrast, corticosterone treatment induced cell death at 250 μM, which led to a 70% reduction in cell viability. However, when cells were treated with SOCG prior to corticosterone treatment, the secretion of LDH (a cytosolic enzyme released from dying cells) decreased and cell viability (determined *via* an MTT assay) increased. Caspase-3 expression was also diminished by SOCG pretreatment, and the ratio of Bax

Table 4

Effect of SOCG on LDH release in corticosterone-treated PC12 cells.

Group	SOCG (μg/mL)	Corticosterone (250 μM)	LDH release (%)
Normal	–	–	100.0 ± 2.8
Control	–	+	150.3 ± 3.6 ^{##}
SOCG	1	+	152.3 ± 1.1
	5	+	141.3 ± 5.4
	10	+	135.4 ± 1.6 [*]
	25	+	124.8 ± 6.0 ^{***}

Values are expressed as means ± SD (n=3).

^{*} p < 0.05 significantly different from corticosterone-treated control group.^{***} p < 0.001 significantly different from corticosterone-treated control group.^{##} p < 0.001 significantly different from normal group.

to Bcl-2 expression decreased, indicating a reduction in cell death. Consistent with *in vivo* data, levels of BDNF mRNA expression significantly increased in corticosterone-treated cells by SOCG. In summary, our data suggested that SOCG is able to control systemic corticosterone production *in vivo* and protect neurons from corticosterone-induced damages *in vitro*. This process might be mediated by interference with the cell death signaling pathway and induction of nerve growth factor expression. Our study demonstrates the need for more detailed investigations of the neuroprotective effects of SOCG using other methods such as the TUNEL assay and evaluation of *in vitro* neurogenesis.

The antidepressant drugs currently available act *via* monoamine neurotransmitters. They include tricyclic antidepressants, which inhibit the plasma membrane transporters for serotonin and/or noradrenaline, monoamine oxidase inhibitors, which reduce the enzymatic breakdown of serotonin and noradrenaline, and newer classes of antidepressant such as the SSRIs (selective serotonin reuptake inhibitors) and SNRIs (serotonin and noradrenaline reuptake inhibitors) (Bouvier et al., 2003). Although these antidepressant drugs are generally safe and effective, only ~50% individuals with depression show full remission. Furthermore, it takes several weeks or even months to receive the clinical benefits of the drugs, and side effects such as mania, restlessness, agitation and insomnia remain a major problem (Berton and Nestler, 2006). Therefore, there is an urgent need for new depression treatments that act more rapidly and are safer and more effective, and that may target different pathologic pathways. Herbal medicine formula usually consists of multiple herbs; therefore, they may be able to target multiple pathways. Here, we suggested that SOCG might protect cells from corticosterone-induced neuronal damage. However, SOCG may have effects on other pathophysiological targets of depression; thus, further research is definitely required using diverse depression models and targets, and further, the active compounds of SOCG should be examined.

Conflict of interest

No competing financial interests exist.

Author contributions

JE Choi, D Park, E Chun, JH Seo, S Kim, and M Do performed experiments and analyzed data. JJ Choi, SY Kim, and Y Park analyzed and discussed data. IC Jung and M Jin designed and supervised research and wrote the paper.

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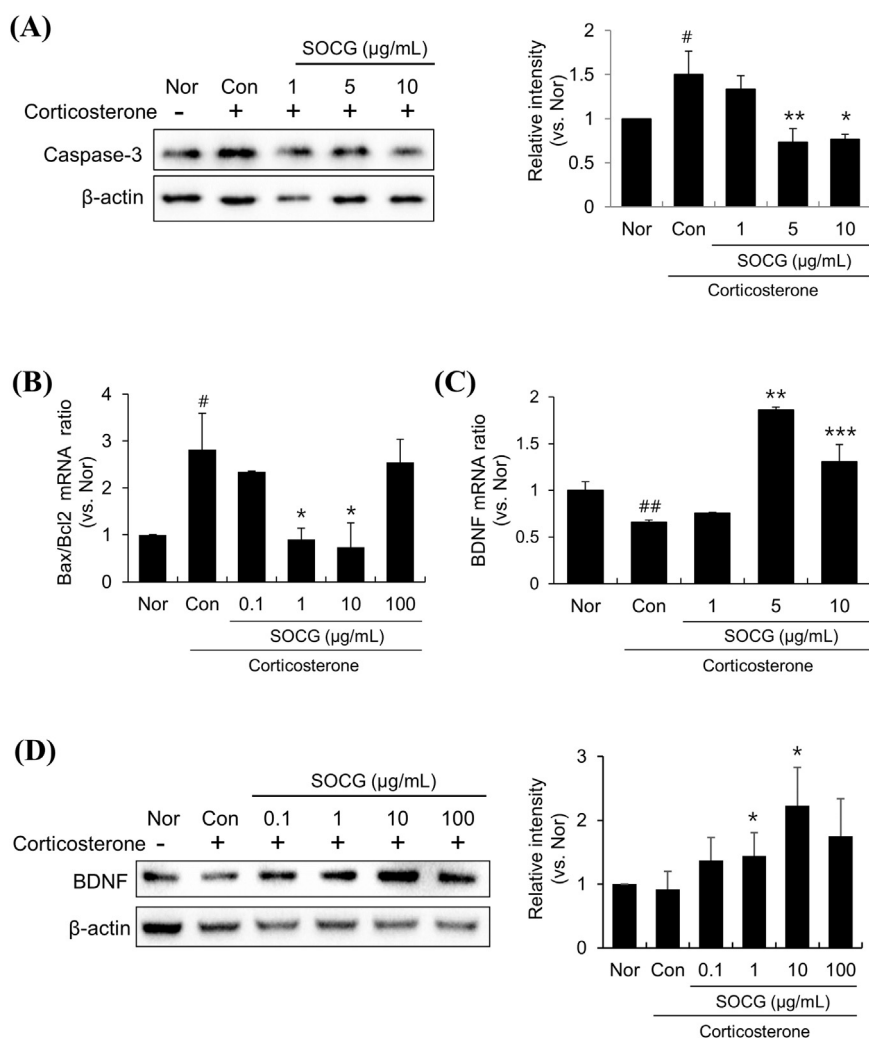


Fig. 5. Effects of SOCG on corticosterone-treated PC12 cells (n=3). (A) Expression of caspase-3, (B) The ratio of Bax/Bcl-2 mRNA expression and (C) BDNF mRNA expression in corticosterone-treated PC12 cells. (D) The cells were treated with SOCG and corticosterone for 24 h. The cell lysates were harvested and subjected to Western blot analysis with BDNF specific antibody. Data are mean $\pm$ SD. [#]p < 0.05, ^{##}p < 0.01 significantly different from normal group. ^{*}p < 0.05, ^{**}p < 0.01, ^{***}p < 0.001 significantly different from corticosterone-treated control group. Nor, normal; Con, control.

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